

# A Unified Framework for Adsorption Isotherm Analysis in Water Quality Studies: Descriptors, Model Selection, and Reporting Guidelines

Eugenio H. Otal,<sup>\*,†,‡,¶</sup> Manuela L. Kim,<sup>§,†</sup> Mutsumi Kimura,<sup>§,‡,||,¶</sup> Katsuya  
Teshima,<sup>¶,†,‡</sup> and Hideki Tanaka<sup>\*,¶,†,‡</sup>

<sup>†</sup>*Department of Materials Chemistry, Faculty of Engineering, Shinshu University, 4-17-1  
Wakasato, Nagano 380-8553, Japan*

<sup>‡</sup>*Research Initiative for Supra-Materials (RISM), Interdisciplinary Cluster for Cutting  
Edge Research (ICCER), Shinshu University, Nagano 380-8553, Japan*

<sup>¶</sup>*Institute for Aqua Regeneration, Shinshu University, 4-17-1 Wakasato, Nagano 380-8553,  
Japan*

<sup>§</sup>*Department of Chemistry and Materials, Faculty of Textile Science and Technology,  
Shinshu University, Ueda 386-8567, Japan*

<sup>||</sup>*COI Aqua-Innovation Center, Shinshu University, Ueda 386-8567, Japan*

E-mail: eugenio\_otal@shinshu-u.ac.jp; htanaka@shinshu-u.ac.jp

Phone: +81 (0)268-21-5499. Fax: +81 (0)268-21-5499

## Abstract

Adsorption isotherms are central to water-quality research, yet their fitted parameters are often compared across models as if they were physically interchangeable. This practice can be misleading because common isotherm equations differ in their asymptotic behavior, parameter dimensions, and assumptions about surface heterogeneity. Here, we propose a unified framework for adsorption isotherm analysis aimed at making capacity, affinity, and heterogeneity descriptors comparable across adsorbents, contaminants, and studies. The framework separates fitted model parameters from reportable physical descriptors and defines a minimum descriptor set consisting of the saturation capacity  $Q_{\max}$ , a dimensionless standard-state affinity  $K_{\text{aff}}^*$ , the corresponding adsorption-energy descriptor  $E_{\text{ads}}$ , and a reported heterogeneity width  $\sigma_H$ . We show that  $Q_{\max}$  should be reported only for models with a finite saturation plateau, whereas non-saturating models such as Freundlich cannot provide a physically meaningful maximum capacity. Model-dependent affinity constants are recast through a declared standard-state concentration  $C^0$ , allowing thermodynamic comparison on a common scale. Surface heterogeneity is treated through the energy distributions implied by each isotherm family. In addition, we provide a practical workflow combining non-linear regression, residual metrics, information criteria, F-tests, physical decision rules, and optional KNN/SVM-based material classification. Together, these guidelines provide a reproducible route from raw isotherm data to statistically validated, physically interpretable, and cross-study comparable adsorption descriptors for water-treatment materials.

## 1 Introduction

Access to safe water is becoming one of the defining challenges of this century. Global population reached approximately 8.2 billion in 2024 and is projected to approach 9.7 billion by 2050.<sup>1</sup> Yet, 2.2 billion people still lack safely managed drinking water services.<sup>2</sup> In addition, roughly half of the world’s population experiences severe water scarcity for at least part of

the year.<sup>3</sup> Urbanization, industrial activity, agricultural intensification, and climate change are compounding these pressures on global water systems.<sup>4</sup> Within this context, the United Nations Sustainable Development Goal 6 (SDG 6), “Clean Water and Sanitation,” has catalyzed broad scientific and policy interest in water treatment technologies. However, recent global assessments indicate that the world is not on track to achieve SDG 6 by 2030.<sup>5</sup>

Among the available remediation strategies, adsorption remains especially attractive because of its operational simplicity, compatibility with decentralized treatment, and applicability across a wide range of contaminant classes. Historically, adsorption research in water purification has focused on metals, metalloids, dyes, and pesticides.<sup>6–8</sup> The contaminant landscape, however, is expanding rapidly and now extends far beyond these classical targets. Contemporary challenges increasingly involve persistent and mobile organic pollutants, pharmaceutical residues, endocrine-disrupting compounds, microplastic-associated chemicals, and, most prominently, per- and polyfluoroalkyl substances (PFAS).<sup>9–12</sup> PFAS have become emblematic of the next generation of adsorption targets: they are highly persistent, often mobile in water, resistant to conventional degradation, and subject to tightening regulatory limits in drinking water worldwide.<sup>13–15</sup> As the diversity of target contaminants grows, so does the need for rigorous, cross-pollutant comparison of adsorbent performance, a task that demands a common descriptive language for adsorption data.<sup>16,17</sup>

Adsorption isotherms are the standard tool for summarizing material performance and for extracting descriptors such as maximum uptake capacity, affinity, and site heterogeneity. In practice, however, parameters obtained from different isotherm models are routinely tabulated side by side as though they were interchangeable. They are not. The Langmuir, double Langmuir, Freundlich, and Sips equations arise from different physical assumptions and assign different meanings, dimensions, and asymptotic behaviors to their fitted constants.<sup>18–20</sup> Comparing a Langmuir  $Q_{\max}$  with a value extrapolated from a Freundlich fit, or placing a Langmuir affinity constant next to a Sips affinity constant whose units depend on the heterogeneity exponent, is neither mathematically consistent nor physically meaning-

55 ful.<sup>21,22</sup> What often appears to be a comparison of materials is, in reality, a comparison of  
56 incompatible parameterizations.<sup>16</sup>

57 This problem is not merely academic. As adsorption studies move toward application-  
58 driven screening of advanced sorbents for complex water matrices—including PFAS, phar-  
59 maceuticals, and other emerging contaminants—materials are increasingly ranked as “better”  
60 on the basis of larger fitted constants derived from different models, concentration scales,  
61 and assumptions about site heterogeneity. Such practice weakens reproducibility, complicates  
62 translation from laboratory results to engineering design, and, most importantly, prevents the  
63 community from building reliable structure–performance relationships across adsorbents and  
64 pollutants. A field that aims to address global water challenges, including those highlighted  
65 by SDG 6 and the rapidly evolving contaminant landscape, requires reporting standards that  
66 make adsorption parameters comparable in a thermodynamically meaningful way.

67 In this work, we propose a practical reporting framework grounded in four principles.  
68 First, capacity comparisons should be restricted to isotherm models that define a finite  
69 saturation capacity. Second, model-dependent affinity constants should be converted, when-  
70 ever possible, into a dimensionless standard-state affinity and the corresponding adsorption-  
71 energy descriptor. Third, model selection should be based on residual diagnostics, informa-  
72 tion criteria, nested-model tests, and physical admissibility, rather than on  $R^2$  alone. Fourth,  
73 once reliable descriptors have been extracted, they can be used for optional downstream clas-  
74 sification or clustering of adsorption systems. The mathematical derivations are provided in  
75 the SI, and the companion notebooks illustrate the workflow with synthetic examples and  
76 user-supplied data.

## 2 The problem: incompatibility of isotherm-derived parameters

### 2.1 Common isotherm models and their assumptions

The most widely used isotherm equations in aqueous-phase adsorption can be written as follows. A complete list is included in Table 1 and SI.

**Langmuir.** Assumes a homogeneous surface with a single class of equivalent sites and monolayer coverage:

$$q_e = \frac{Q_{\max} K_L C_e}{1 + K_L C_e}. \quad (1)$$

Parameters:  $Q_{\max}$  ( $\text{mg g}^{-1}$ ), the finite saturation capacity, and  $K_L$  ( $\text{L mg}^{-1}$ ), the affinity constant. See Eq. (S-3).

**Double Langmuir.** Extends Langmuir to two independent site populations:

$$q_e = \frac{Q_1 K_1 C_e}{1 + K_1 C_e} + \frac{Q_2 K_2 C_e}{1 + K_2 C_e}. \quad (2)$$

Total capacity  $Q_{\max} = Q_1 + Q_2$  remains finite and well defined. See Eq. (S-14).

**Freundlich.** An empirical power law that describes heterogeneous adsorption without a saturation plateau:

$$q_e = K_F C_e^{1/n}. \quad (3)$$

$K_F$  has concentration-dependent units  $(\text{mg g}^{-1})(\text{L mg}^{-1})^{1/n}$ ; no  $Q_{\max}$  is defined. See Eq. (S-22).

91 **Sips (Langmuir–Freundlich).** Introduces surface heterogeneity through an exponent  $m$   
 92 while retaining a saturation plateau:

$$q_e = \frac{Q_{\max} K_S C_e^m}{1 + K_S C_e^m}. \quad (4)$$

93 When  $m = 1$ , Sips reduces to Langmuir. The affinity constant  $K_S$  carries units that depend  
 94 on  $m$ , i.e.  $(\text{L mg}^{-1})^m$ , and is therefore not directly comparable with  $K_L$ . See Eq. (S-29).

**Parameterization note.** In this manuscript, the Sips constant is defined outside the concentration exponent,  $K_S C_e^m$ , consistently in the main text, SI, and fitting code. With this convention  $K_S$  carries units of  $[C]^{-m}$ , and the reciprocal-concentration affinity used for descriptor extraction is  $K_S^{1/m}$ .

95 As shown in Figures 1–3, the model selection procedure combines physical screening,  
 96 statistical comparison, and physicochemical validation.

Table 1: Availability of  $Q_{\max}$ ,  $K_H$ , and an effective affinity constant  $K_{\text{aff}}$  for common isotherm models, together with their implied energy-distribution family. The “Recommended use” column is **YES** when both  $Q_{\max}$  and a comparable  $K_{\text{aff}}$  can be reported, and **NO** otherwise. See the SI sections in the last column for the derivations and definitions.

| Model                | $Q_{\max}$ | $K_H$ | $K_{\text{aff}}$ | Recommended | Energy distribution    | SI ref. |
|----------------------|------------|-------|------------------|-------------|------------------------|---------|
| Henry                | No         | Yes   | No               | NO          | – (low-coverage limit) | S-1.1   |
| Langmuir             | Yes        | Yes   | Yes              | YES         | Dirac $\delta$         | S-1.2   |
| Double<br>Langmuir   | Yes        | Yes   | Yes              | YES         | Two Dirac $\delta$     | S-1.3   |
| Freundlich           | No         | Cond. | No               | NO          | Power law              | S-1.4   |
| Sips (L–F)           | Yes        | Cond. | Yes *            | YES         | Logistic               | S-1.5   |
| Tóth                 | Yes        | Yes   | Yes              | YES         | Gen. logistic (asym.)  | S-1.6   |
| Jovanovic            | Yes        | Yes   | Yes              | YES         | Gumbel (EV-I)          | S-1.7   |
| Temkin               | No         | No    | Yes *            | NO          | Uniform (rect.)        | S-1.8   |
| UNILAN               | Yes        | Yes * | Yes              | YES         | Uniform (rect.)        | S-1.9   |
| Redlich–<br>Peterson | Cond.      | Yes   | Yes *            | NO          | Non-standard (asym.)   | S-1.10  |
| Koble–<br>Corrigan   | Yes        | Cond. | Yes *            | YES         | Logistic (Sips like)   | S-1.11  |
| Radke–<br>Prausnitz  | No         | Yes   | Cond. *          | NO          | Non-standard (asym.)   | S-1.12  |
| D-R / D-A            | Yes        | No    | Yes *            | YES         | Rayleigh / Weibull     | S-1.13  |
| Hill                 | Yes        | Cond. | Yes *            | YES         | Logistic (Sips like)   | S-1.14  |

**Notes.** “Cond.” indicates availability only in special parameter limits (e.g.,  $m = 1$ ,  $n_H = 1$ ,  $\beta = 1$ ). “\*” indicates an effective or operational definition; see the SI for explicit formulas and limits.

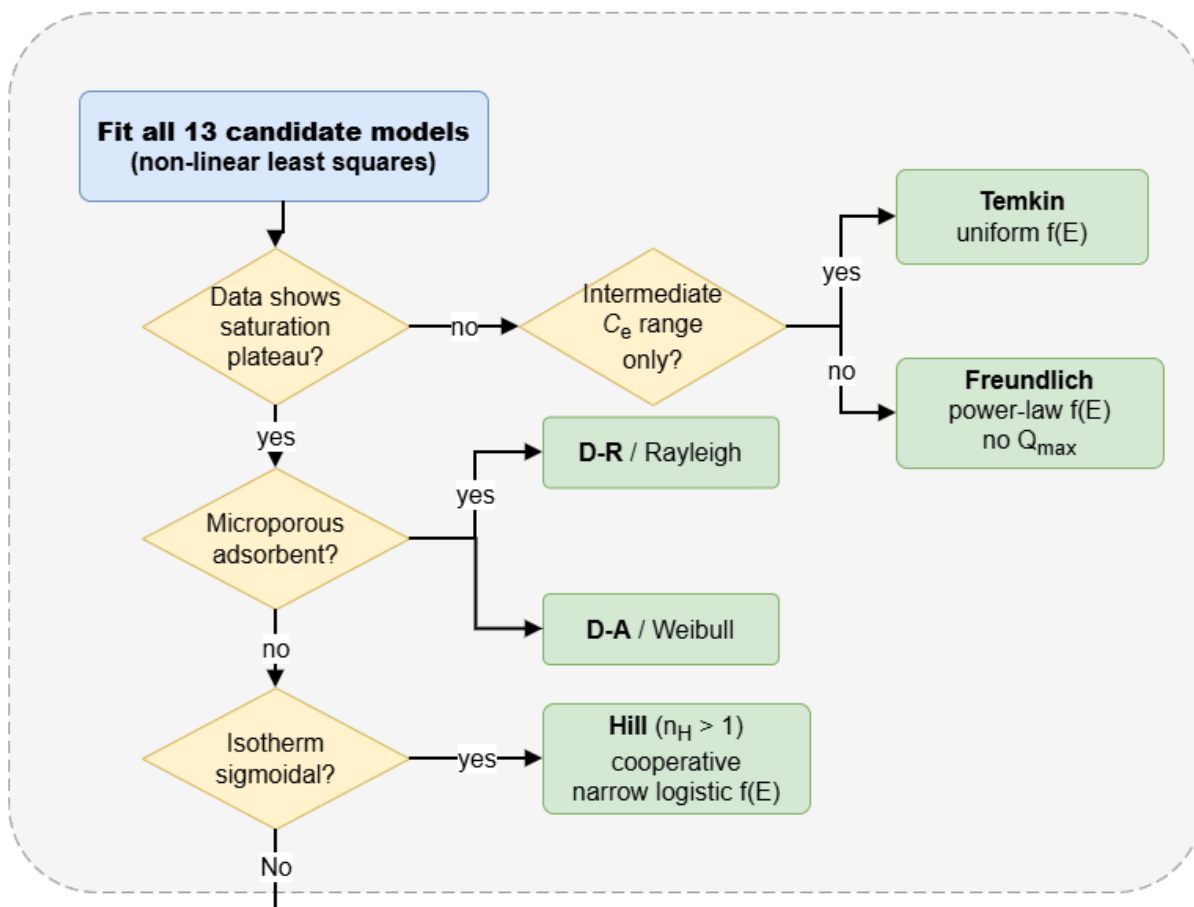


Figure 1: Shape-based pre-screening of adsorption isotherm models. Experimental isotherms are first fitted with the diagnostic model library by non-linear least squares and then classified according to physically observable features, including the presence of a saturation plateau, microporous uptake behavior, sigmoidal adsorption, and restriction to an intermediate concentration range. This first decision layer separates models that can provide finite capacity descriptors from models that are better interpreted as local, empirical, or special-regime descriptions.



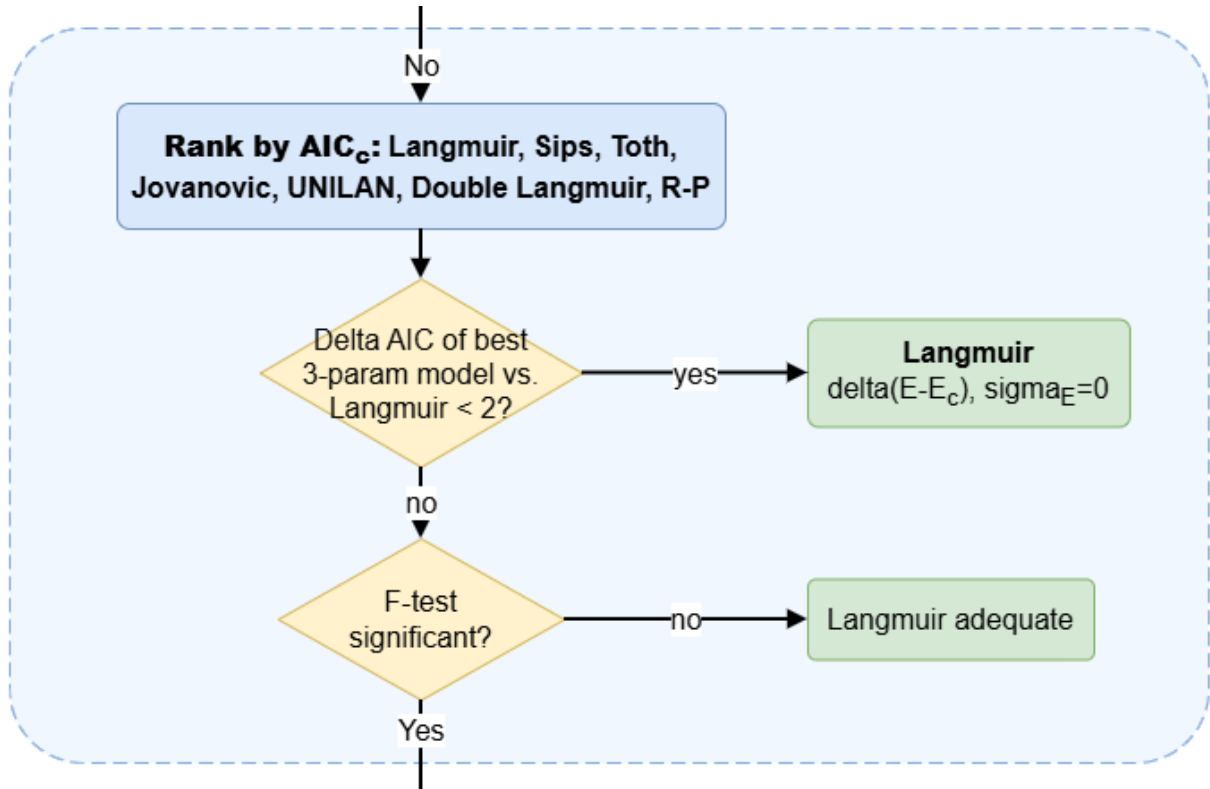


Figure 2: Statistical model-selection step used to decide whether a model more complex than Langmuir is justified. Candidate saturating models are ranked by the corrected Akaike information criterion ( $AIC_c$ ), and the best three-parameter model is compared against Langmuir. When the improvement in  $AIC_c$  is small or the F-test is not significant, the Langmuir model is retained as an adequate description, avoiding over-interpretation of weak heterogeneity.

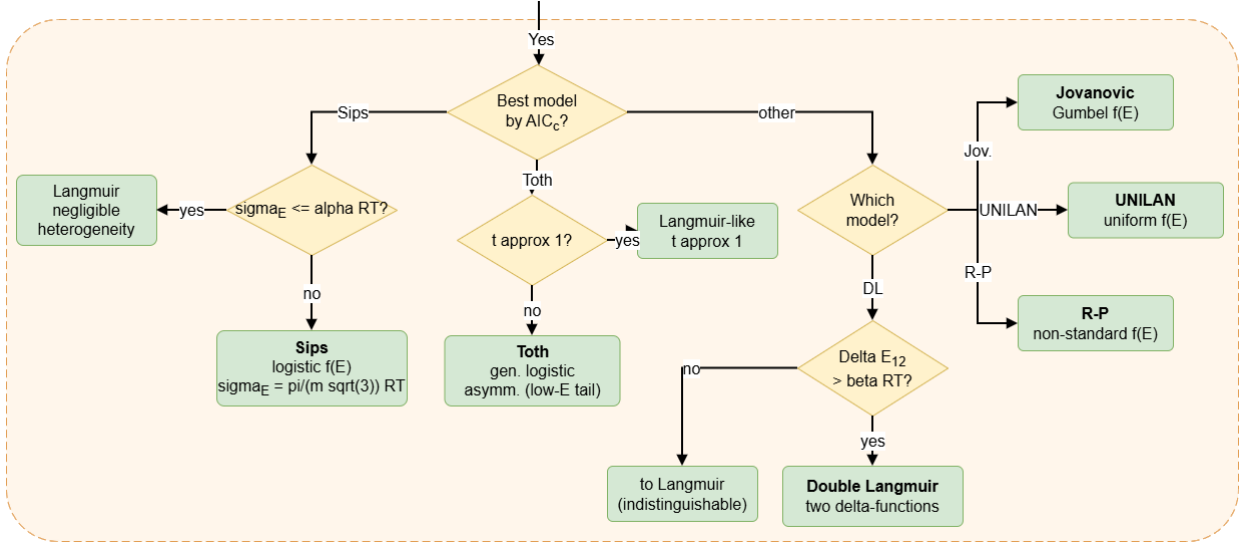


Figure 3: Final physical-validation step for assigning the selected isotherm model to a reportable adsorption descriptor set. The best  $AIC_c$  model is interpreted in terms of its implied energy-distribution family and limiting behavior. Sips and  $T\tilde{A}^3$  fits are checked for proximity to the Langmuir limit, while Jovanovic, UNILAN, Redlich–Peterson, and double-Langmuir outcomes are assigned according to their characteristic heterogeneity signatures. This step links statistical goodness of fit with physically meaningful descriptors such as finite capacity, affinity, and energy-distribution width.

## 2.2 Why $Q_{\max}$ values are not always comparable

Even though  $Q_{\max}$  is often treated as a universal figure of merit, it is model-specific unless the isotherm explicitly contains a finite saturation plateau. In the Langmuir and Double Langmuir equations,  $Q_{\max}$  (or  $Q_1 + Q_2$ ) is a true asymptote: it is the coverage reached as  $C_e \rightarrow \infty$  and is therefore a well-defined thermodynamic capacity. By contrast, the Freundlich isotherm has no saturation term;  $q_e$  increases without bound as  $C_e$  increases (SI Eq. (S-22)), so any “ $Q_{\max}$ ” inferred from a Freundlich fit depends entirely on the chosen concentration window. Reporting such a value as a capacity is physically meaningless because it is not a model parameter and has no invariant interpretation.

Even for models that do saturate, comparability depends on whether the saturation regime is actually constrained by the experimental data. In the Sips equation, for example, the heterogeneity exponent  $m$  can shift the onset of saturation far beyond the measured

concentration range (see Eq. S-29). Under these conditions, the fitted  $Q_{\max}$  is not determined by the data but inferred through extrapolation. As a result, reported capacities may reflect model-imposed asymptotes rather than experimentally supported saturation values. The practical consequence is that tables mixing Langmuir or Double Langmuir  $Q_{\max}$  values with Freundlich-derived or poorly constrained Sips capacities are not comparing the same physical quantity. Any ranking built on such a mixture conflates experimentally supported saturation capacities with model-dependent extrapolations.

## 2.3 Why affinity constants are not comparable across models

Affinity constants carry the strongest illusion of comparability because they are reported as single numbers, yet their physical meaning and units change from model to model. In the Langmuir equation,  $K_L$  has units of inverse concentration (e.g.,  $\text{L mg}^{-1}$ ) and sets the midpoint of the uptake curve. In the Sips equation, the constant  $K_S$  carries units of  $(\text{L mg}^{-1})^m$ , which depend on the heterogeneity exponent  $m$ ; therefore its numerical value is not on the same scale as  $K_L$  unless  $m = 1$  (SI Eqs. (S-3) and (S-29)). The Freundlich constant  $K_F$  is even less comparable, because its units depend on  $1/n$  and it does not correspond to any saturation behavior (SI Eq. (S-22)).

The Henry constant  $K_H$  provides a low-coverage slope (SI Eq. (S-7)), but it is not a full-isotherm descriptor and cannot replace affinity constants derived from different functional forms without additional normalization. As a result, placing  $K_L$ ,  $K_S$ , and  $K_F$  side by side implies a common meaning that does not exist. Any comparison based on raw affinity constants is therefore a comparison of incommensurate parameters, not of intrinsic adsorption strength.

### 3 A common language: standard-state adsorption energy

#### 3.1 Constructing a dimensionless equilibrium constant

To place all isotherm models on a common thermodynamic footing, the model-specific affinity parameter must be expressed as a dimensionless equilibrium constant. This is necessary because the fitted affinity constants used in adsorption isotherms are usually dimensional quantities whose units depend on the mathematical form of the model and on the units chosen for concentration and uptake. Therefore, direct comparison of fitted affinity parameters between different models can be misleading unless a common normalization procedure is applied.

Here, the affinity is defined from the low-concentration Henry region of the isotherm. The Henry constant is obtained as the initial slope,

$$K_H = \lim_{C_e \rightarrow 0} \left( \frac{q_e}{C_e} \right), \quad (5)$$

where  $q_e$  is the equilibrium uptake and  $C_e$  is the equilibrium concentration in solution. For example, if  $q_e$  is expressed in  $\text{mg g}^{-1}$  and  $C_e$  in  $\text{mg L}^{-1}$ , then

$$[K_H] = \frac{\text{mg g}^{-1}}{\text{mg L}^{-1}} = \text{L g}^{-1}. \quad (6)$$

Since  $K_H$  still contains the adsorbent mass dependence, it is normalized by the maximum adsorption capacity  $Q_{\max}$ . If  $Q_{\max}$  is expressed in  $\text{mg g}^{-1}$ , then

$$\left[ \frac{K_H}{Q_{\max}} \right] = \frac{\text{L g}^{-1}}{\text{mg g}^{-1}} = \text{L mg}^{-1}. \quad (7)$$

Thus,  $K_H/Q_{\max}$  has units of inverse concentration. A dimensionless affinity constant is then obtained by multiplying this quantity by a declared standard-state concentration  $C^0$

expressed in the same concentration units as  $C_e$ :

$$K_{\text{aff}}^* = \frac{K_H}{Q_{\text{max}}} C^0. \quad (8)$$

For instance, if  $C^0$  is expressed in  $\text{mg L}^{-1}$ , then

$$[K_{\text{aff}}^*] = (\text{L mg}^{-1}) (\text{mg L}^{-1}) = 1. \quad (9)$$

Therefore,  $K_{\text{aff}}^*$  is dimensionless by construction. For comparisons among different adsorbates, the declared standard state should preferably be expressed on a molar concentration scale or, when available, in terms of activity. Concentrations in  $\text{mg L}^{-1}$  are acceptable for water-quality reporting and engineering screening, but they should be interpreted as operational descriptors rather than as composition-independent thermodynamic standard states. For models such as Sips or Koble–Corrigan, where the initial slope at  $C_e \rightarrow 0$  diverges (i.e., a finite Henry constant does not exist), an operational affinity constant is defined instead by raising the native affinity parameter (carrying fractional units) to the power of the reciprocal exponent (e.g.,  $K_S^{1/m}$ ), thereby restoring the units of inverse concentration before normalization.

In this construction,  $Q_{\text{max}}$  removes the dependence of the Henry slope on the total adsorption capacity, while  $C^0$  removes the remaining concentration unit. Thus,  $K_{\text{aff}}^*$  represents the standard-state-normalized affinity of the adsorbent–adsorbate pair relative to a defined concentration standard. For the Langmuir isotherm, for example,  $K_H = Q_{\text{max}} K_L$ , and therefore  $K_{\text{aff}}^* = K_L C^0$ , recovering the usual dimensionless form of the Langmuir equilibrium constant. The same procedure can be applied to any isotherm model for which a finite Henry limit and a meaningful saturation capacity can be defined.

It should be emphasized that the numerical value of  $K_{\text{aff}}^*$  depends on the chosen standard state  $C^0$ . Therefore,  $C^0$  must be explicitly reported. However, once the same  $C^0$  is used consistently, the resulting dimensionless constants allow affinity parameters extracted

173 from different isotherm models to be compared on the same basis and used in standard  
 174 thermodynamic expressions.

## 175 **3.2 Standard-state adsorption energy $E_{\text{ads}}$**

176 The standard Gibbs free energy associated with the declared standard state is

$$\Delta G_{\text{ads}}^0 = -RT \ln K_{\text{aff}}^*. \quad (10)$$

177 For reporting, we use the positive affinity-energy descriptor

$$E_{\text{ads}} \equiv E_{\text{aff}} = -\Delta G_{\text{ads}}^0 = RT \ln K_{\text{aff}}^*. \quad (11)$$

178 Thus, larger  $E_{\text{ads}}$  corresponds to stronger standard-state affinity. Because  $K_{\text{aff}}^*$  is dimension-  
 179 less, the logarithmic term is dimensionless and the units of  $E_{\text{ads}}$  arise only from  $RT$ :

$$[E_{\text{ads}}] = [RT] = \text{kJ mol}^{-1} \quad (12)$$

180 when  $R$  is expressed in  $\text{kJ mol}^{-1} \text{K}^{-1}$ . The descriptor is therefore a standardized model-  
 181 derived measure of affinity, not a raw fitted parameter and not a directly measured observable  
 182 independent of the fitted model.

## 183 **3.3 Extraction of $K_{\text{aff}}^*$ and $E_{\text{ads}}$ from each model**

184 Table 2 shows how  $K_{\text{aff}}^*$  and  $E_{\text{ads}}$  are obtained from the native parameters of each isotherm.

Table 2: Extraction of native and operational affinity descriptors from isotherm parameters.  $C^0$  is the declared standard-state concentration; descriptors are reportable only when the fitted model is physically admissible and identifiable.

| Model            | $Q_{\max}$                   | $K_{\text{aff}}$ (native)                                  | $E_{\text{ads}}/(RT)$                        |
|------------------|------------------------------|------------------------------------------------------------|----------------------------------------------|
| Langmuir         | $Q_{\max}$                   | $K_L$                                                      | $\ln(K_L C^0)$                               |
| Double Langmuir  | $Q_1 + Q_2$                  | $K_L^{(\text{tot})} = \frac{Q_1 K_1 + Q_2 K_2}{Q_1 + Q_2}$ | $\ln(K_L^{(\text{tot})} C^0)$                |
| Sips             | $Q_{\max}$                   | $K_S^{1/m}$                                                | $\frac{1}{m} \ln(K_S (C^0)^m)$               |
| Tóth             | $q_{\max}$                   | $b$                                                        | $\ln(b C^0)$                                 |
| Jovanovic        | $q_{\max}$                   | $b$                                                        | $\ln(b C^0)$                                 |
| Temkin           | —                            | $a_T$                                                      | $\ln(a_T C^0)$                               |
| UNILAN           | $Q_{\max}$                   | $\sqrt{b_{\max} b_{\min}}$                                 | $\frac{1}{2} \ln(b_{\max} b_{\min} (C^0)^2)$ |
| Redlich–Peterson | $K_{RP}/a_{RP}$ <sup>a</sup> | $a_{RP}$ <sup>a</sup>                                      | $\ln(a_{RP} C^0)$                            |
| Koble–Corrigan   | $A_{KC}/B_{KC}$              | $B_{KC}^n$                                                 | $n \ln(B_{KC} C^0)$                          |
| Radke–Prausnitz  | —                            | $r_{RP}$                                                   | $\ln(r_{RP} C^0)$                            |
| D-R / D-A        | $Q_{\max}$                   | $1/C_{1/2}$ <sup>b</sup>                                   | $\ln(C^0/C_{1/2})$                           |
| Hill             | $Q_{\max}$                   | $K_D^{-1/n_H}$                                             | $-\frac{1}{n_H} \ln K_D + \ln C^0$           |

<sup>a</sup> Only when  $\beta = 1$ . <sup>b</sup>  $C_{1/2}$  is the half-loading concentration; see Sec. S-1.13. “—” indicates no finite  $Q_{\max}$  exists.

185 The Freundlich isotherm may describe adsorption data over a restricted concentration  
186 window, but it is absent from Table 2 because it defines no finite  $Q_{\max}$ . Consequently,  
187 the ratio  $K_H/Q_{\max}$  is undefined and no dimensionless affinity can be constructed from the  
188 Freundlich parameters alone. For the Double Langmuir model, an effective total adsorption  
189 energy can be reported from the capacity-weighted constant  $K_L^{(\text{tot})}$ , while the site-resolved  
190 constants  $K_1$  and  $K_2$  remain necessary to preserve the bimodal interpretation captured by  
191 the site-resolved energy separation. Finally, the choice of standard-state concentration  $C^0$   
192 must be explicitly declared; although it appears inside the logarithm and therefore cancels in  
193 any *difference*  $\Delta E_{\text{ads}}$ , it must be consistent across all entries in a comparative table. The full  
194 set of energy-distribution families and their relation to  $\sigma_E$  is given in Table S-1; numerical  
195 values of  $\sigma_E$  at  $T = 298$  K are collected in Table S-2.

## 4 Quantifying surface heterogeneity: energy distributions

### 4.1 From adsorption sites to observable energy distributions

The final variable in the adsorption system is the distribution of site energies. Unlike  $Q_{\max}$  or  $E_{\text{ads}}$ , this quantity is usually not fitted explicitly, yet it is implicitly encoded in every isotherm model. In other words, each adsorption equation corresponds to a particular assumption about how adsorption energies are distributed across the surface. This provides a natural way to place different isotherm models in a common physical framework.

Real adsorbent surfaces are energetically heterogeneous because adsorption sites differ in local chemistry, accessible geometry, coordination environment, and confinement. As a result, adsorption is more realistically described by a distribution of site energies than by a single characteristic value. A heterogeneous surface can therefore be represented as an ensemble of independent Langmuir-type sites, each characterized by an adsorption energy  $E$ , so that the macroscopic isotherm is written as

$$q_e = Q_{\max} \int_{-\infty}^{\infty} \Theta_L(E - E_c) f(E) dE, \quad (13)$$

where  $f(E)$  is the normalized distribution of adsorption site energies,

$$\int_{-\infty}^{\infty} f(E) dE = 1, \quad (14)$$

and  $\Theta_L(E - E_c)$  is the Langmuir occupation kernel,

$$\Theta_L(E - E_c) = \frac{1}{1 + \exp[-(E - E_c)/(RT)]}. \quad (15)$$



211 Here  $E_c$  is the condensation energy associated with the equilibrium concentration,

$$E_c = RT \ln \left( \frac{1}{K_0 C_e} \right), \quad (16)$$

212 with  $K_0$  a reference constant used to define the energy scale.

213 Equation (13) emphasizes an important point: the experimentally observed isotherm  
 214 does not probe the intrinsic distribution  $f(E)$  directly, but rather the convolution of that  
 215 distribution with the thermal occupation kernel. Differentiating eq (13) with respect to  $E_c$   
 216 gives the observable energy-density profile

$$\phi(E_c) = -\frac{1}{Q_{\max}} \frac{dq_e}{dE_c} = \int_{-\infty}^{\infty} f(E) g_T(E - E_c) dE, \quad (17)$$

217 where

$$g_T(x) = \frac{\exp(-x/RT)}{RT [1 + \exp(-x/RT)]^2} \quad (18)$$

218 is the thermal broadening kernel. This kernel is a logistic probability density with standard  
 219 deviation

$$\sigma_T = \frac{\pi}{\sqrt{3}} RT. \quad (19)$$

220 Thus, even a perfectly homogeneous Langmuir surface, intrinsically described by a Dirac  
 221 distribution,

$$f(E) = \delta(E - E_1), \quad (20)$$

222 is observed through the broadened profile

$$\phi(E_c) = g_T(E_1 - E_c), \quad (21)$$

223 which already has a finite width set by temperature. At  $T = 298$  K, this intrinsic thermal  
 224 width is

$$\sigma_T \approx 4.5 \text{ kJ mol}^{-1}. \quad (22)$$

225 This provides a physical interpretation for why systems with very small energetic dispersion  
 226 often appear experimentally indistinguishable from Langmuir behavior.

227 The commonly used condensation approximation replaces the smooth Langmuir kernel  
 228 by a step function,

$$\Theta_L(E - E_c) \approx \Theta(E - E_c), \quad (23)$$

229 so that thermal broadening is neglected and the intrinsic distribution can be recovered di-  
 230 rectly as

$$f(E) = -\frac{1}{Q_{\max}} \frac{dq}{dE}. \quad (24)$$

231 This approximation is useful because it provides a practical route from any empirical isotherm  
 232 equation back to its implied energy distribution. Full derivations for the models treated here  
 233 are given in Section S-4.4.

## 234 **4.2 Discrete sites, thermal broadening, and continuous heterogene-** 235 **ity**

236 The relation between intrinsic and observable distributions clarifies how the usual isotherm  
 237 families are connected. Consider first a surface with two discrete classes of sites,

$$f(E) = w \delta(E - E_1) + (1 - w) \delta(E - E_2), \quad 0 \leq w \leq 1. \quad (25)$$

238 Under thermal broadening, the experimentally accessible profile becomes

$$\phi(E_c) = w g_T(E_1 - E_c) + (1 - w) g_T(E_2 - E_c). \quad (26)$$

239 If the two site energies are sufficiently separated relative to the thermal broadening scale,  
 240 the bimodal character is preserved and a Double Langmuir description remains physically  
 241 meaningful. For the symmetric case of equal populations, the onset of bimodality for the

242 logistic kernel occurs at

$$|E_1 - E_2| > 2RT \operatorname{arcosh}(2) \approx 2.63 RT \approx 1.45 \sigma_T. \quad (27)$$

243 Below this scale, the two thermally broadened contributions overlap strongly and may appear  
244 as a single broadened population. In that regime, the same data may be described by a  
245 continuous heterogeneous model such as Sips or Tóth rather than by two explicitly resolved  
246 Langmuir populations. For unequal populations, the exact threshold also depends on the  
247 statistical weights of the two site classes.

248 This interpretation also explains the role of temperature. Increasing  $T$  increases the  
249 kernel width  $\sigma_T$  according to eq (19), making nearby site populations less distinguishable.  
250 Conversely, lower temperatures sharpen the energetic resolution and make multimodal struc-  
251 ture easier to detect. Therefore, the observed heterogeneity is not only a property of the  
252 surface itself, but also of the thermal scale at which the experiment is performed.

253 In this sense, the standard isotherm families can be viewed as limiting cases of the same  
254 general picture. Langmuir corresponds to a single discrete energy level. Double Langmuir  
255 corresponds to two discrete levels. Sips, Hill, and Koble–Corrigan correspond to symmetric  
256 continuous distributions with finite width. Tóth and Jovanovic correspond to asymmet-  
257 ric continuous distributions. Freundlich represents the limiting case of extremely broad or  
258 effectively unbounded heterogeneity. The different models are therefore not disconnected em-  
259 pirical formulas, but alternative representations of how adsorption energies are distributed  
260 and thermally resolved.

### 261 **4.3 Energy-distribution families implied by common isotherms**

262 Applying eq (24) to each isotherm yields the energy-distribution families summarized in  
263 Table 3. These distributions should be interpreted as the intrinsic site-energy distributions  
264 implied by each model under the condensation approximation.

Table 3: Energy-distribution families and apparent mathematical width  $\sigma_E$  implied by each isotherm model. Parameter  $m$  (Sips, Koble–Corrigan) is the heterogeneity exponent;  $t$  (Tóth) and  $n_H$  (Hill) are the respective shape parameters.  $\sigma_E$  is in units of  $RT$  ( $RT \approx 2.48 \text{ kJ mol}^{-1}$  at 298 K). Full derivations are given in Section S-4.4.

| Model                | Distribution $f(E)$             | Symm. | $\sigma_E/(RT)$                                                  |
|----------------------|---------------------------------|-------|------------------------------------------------------------------|
| Langmuir             | Dirac $\delta(E - E_c)$         | —     | 0                                                                |
| Double Langmuir      | Two Dirac $\delta$              | —     | $\frac{\sqrt{Q_1 Q_2}}{Q_1 + Q_2} \frac{ \Delta E_{12} }{RT}$    |
| Sips                 | Logistic (scale $RT/m$ )        | Yes   | $\pi/(m\sqrt{3})$                                                |
| Tóth                 | Generalized logistic type IV    | No    | $> \pi/\sqrt{3}$ ( $t < 1$ ); no closed form                     |
| Jovanovic            | Gumbel (EV-I)                   | No    | $\pi/\sqrt{6} \approx 1.28$                                      |
| Temkin               | Uniform                         | Yes   | $b_T Q_{\max}/(2\sqrt{3} RT)$                                    |
| UNILAN               | Uniform (exact kernel)          | Yes   | $\ln(b_{\max}/b_{\min})/(2\sqrt{3})$                             |
| Freundlich           | Power law (unbounded)           | —     | $\infty$                                                         |
| Redlich–Peterson     | Non-standard ( $\beta < 1$ )    | No    | —                                                                |
| Koble–Corrigan       | $\equiv$ Sips                   | Yes   | $\pi/(m\sqrt{3})$                                                |
| Radke–Prausnitz      | Non-standard                    | No    | —                                                                |
| D-R ( $n_{DA} = 2$ ) | Rayleigh                        | No    | $0.463 E_a/(RT)$                                                 |
| D-A (general)        | Weibull (shape $n_{DA}$ )       | No    | $(E_a/RT)\sqrt{\Gamma(1 + 2/n_{DA}) - [\Gamma(1 + 1/n_{DA})]^2}$ |
| Hill                 | $\equiv$ Sips (scale $RT/n_H$ ) | Yes   | $\pi/(n_H\sqrt{3})$                                              |

Several physically meaningful relations follow from Table 3. First, Langmuir and Double Langmuir represent discrete energetic populations, whereas Sips, Hill, Koble–Corrigan, Tóth, Jovanovic, Temkin, and UNILAN represent continuous distributions. Second, Sips, Hill, and Koble–Corrigan belong to the same symmetric logistic family and differ only in parameterization. Third, Tóth and Jovanovic introduce skewness, showing that heterogeneity is not fully described by width alone. Fourth, Freundlich corresponds to a limiting case in which the energy dispersion becomes effectively unbounded, consistent with the absence of a finite saturation capacity. Finally, the Dubinin family is naturally expressed in the Polanyi adsorption potential rather than directly in  $E$ , but still admits an equivalent probabilistic interpretation through Weibull-type distributions.

## 4.4 Physical and statistical interpretation of the energy distribution

Once an isotherm has been mapped onto an energy distribution, two widths must be kept conceptually separate. The quantity  $\sigma_E$  denotes the apparent mathematical width of the model-implied energy distribution. The reported heterogeneity descriptor is  $\sigma_H$ , which removes the homogeneous thermal contribution whenever that correction is defined. Therefore, Langmuir is recovered at  $\sigma_H = 0$ , even though thermally broadened occupation kernels may have a finite apparent width. Representative numerical values and the model-specific corrections are given in Table S-2.

This representation also provides a natural connection to statistical model comparison. Two different isotherm models may fit the same data well because their implied energy distributions have similar central energies and similar widths, even if their algebraic forms differ. For example, a narrow bimodal distribution and a single broader logistic distribution may generate nearly indistinguishable isotherms when the two discrete energies are closer than the thermal resolution. Conversely, two models with similar residual error but markedly different  $E_{\text{ads}}$  or  $\sigma_E$  imply different physical pictures of the surface.

The similarity between two implied energy distributions,  $f_1(E)$  and  $f_2(E)$ , can be quantified directly through an overlap coefficient,

$$\text{OVL} = \int_{-\infty}^{\infty} \min[f_1(E), f_2(E)] dE, \quad 0 \leq \text{OVL} \leq 1. \quad (28)$$

High overlap indicates that two models encode essentially the same energetic picture, whereas low overlap indicates physically distinct heterogeneity patterns.

More broadly, the descriptor set

$$\mathbf{d} = \{Q_{\text{max}}, E_{\text{ads}}, \sigma_H\} \quad (29)$$

defines a compact, physically interpretable feature space for adsorption systems. Within this

space,  $Q_{\max}$  measures the amount of adsorption,  $E_{\text{ads}}$  measures the central energetic level, and  $\sigma_H$  measures intrinsic heterogeneity. These descriptors can be used not only to compare fitted models, but also to compare materials, contaminants, or operating conditions on a common basis. They also provide suitable input variables for data-driven analysis, including clustering and supervised classification methods such as k-nearest neighbours (KNN) and support vector machines (SVM), as developed in the following section. In this way, the energy-distribution view links isotherm physics, statistical model discrimination, and descriptor-based classification within a single framework.

## 5 Model selection and statistical comparison

### 5.1 Non-linear regression

To extract the proposed universal descriptors, the diagnostic model library should be fitted to the experimental  $q_e$ - $C_e$  data by non-linear or weighted non-linear regression (e.g. Levenberg–Marquardt or trust-region algorithms), minimising the sum of squared residuals  $\text{SSE} = \sum_i (q_i^{\text{exp}} - q_i^{\text{pred}})^2$  when errors are homoscedastic. When experimental uncertainties or non-constant variance are available, weighted non-linear regression should be used with the corresponding likelihood. Linearised forms (e.g.  $1/q_e$  vs.  $1/C_e$  for Langmuir,  $\ln q_e$  vs.  $\ln C_e$  for Freundlich) must be avoided because they distort the error structure and introduce bias in the estimated parameters. Physical bounds should be imposed on all parameters ( $Q_{\max} > 0$ ,  $K > 0$ ,  $0 < m \leq 1$  for Sips, etc.), and multiple initial guesses should be tested to guard against local minima. Numerical convergence alone is not considered model acceptance. The full model library is used as a diagnostic screen; final interpretation is restricted to models that are physically admissible, identifiable from the available data, and statistically plausible.

Table 4: Model status used in the diagnostic fitting workflow.

| Status       | Meaning                                                      | Used for reporting? |
|--------------|--------------------------------------------------------------|---------------------|
| Fitted       | Numerical convergence was achieved                           | No                  |
| Admissible   | Parameters and limits are physically meaningful              | Maybe               |
| Identifiable | Parameters are constrained by the data and bootstrap-stable  | Maybe               |
| Plausible    | Within statistical support, e.g. $\Delta\text{AIC}_c \leq 2$ | Yes, as candidate   |
| Reported     | Simplest physically meaningful model or descriptor range     | Yes                 |

### 5.1.1 Residual error metrics

No single error metric captures all aspects of fit quality; therefore, multiple complementary metrics should be reported. The complete equations are given in Section S-2.1; here we summarise their purpose and interpretation.

- **SSE** (sum of squared errors, Eq. S-106): the objective function minimised during fitting. Sensitive to large deviations (outliers).
- $R^2$  (coefficient of determination, Eq. S-107): fraction of variance explained. Values close to 1 indicate a good fit, but  $R^2$  alone cannot distinguish overfitting.
- $R^2_{\text{adj}}$  (adjusted  $R^2$ , Eq. S-108): penalises model complexity by accounting for the number of parameters  $p$ . Preferred over  $R^2$  when comparing models with different numbers of parameters.
- **EABS** (sum of absolute errors, Eq. S-109): less sensitive to outliers than SSE. Useful as a complementary absolute-scale measure.
- **RESID** (mean absolute residual, Eq. S-110): average deviation per data point in the same units as  $q_e$ . Provides an intuitive measure of typical prediction error.
- **ARED** (average relative error in percent, Eq. S-111): normalises each residual by  $q_i^{\text{exp}}$ , giving equal weight to low- and high-concentration points. Particularly informative for data spanning several orders of magnitude.

- **MPSED** (median percentage squared error, Eq. S-112): a robust (median-based) relative error metric, resistant to individual outliers.

- **HYBRID** (Eq. S-113): a compromise between SSE and relative error that penalises underfitting at low concentrations, where  $q_i^{\text{exp}}$  is small and squared residuals are amplified by the  $1/q_i^{\text{exp}}$  weight. It also corrects for degrees of freedom ( $n - p$ ).

We recommend reporting at least SSE,  $R_{\text{adj}}^2$ , ARED, and HYBRID for every model. The combination provides sensitivity to both absolute and relative deviations, robustness to outliers, and a degrees-of-freedom correction.

## 5.2 Information criteria: AIC and BIC

In adsorption studies, model selection is often based on goodness-of-fit metrics such as the coefficient of determination ( $R^2$ ) or the sum of squared errors (SSE). However, these metrics systematically favor models with a larger number of adjustable parameters, even when the improvement in fit is not physically meaningful. This is particularly problematic when comparing isotherm models of different complexity, such as Langmuir, Sips, or multi-site formulations, where additional parameters can artificially improve the apparent agreement with experimental data.

To address this issue, information criteria provide a principled framework for balancing model fidelity and complexity. These criteria penalize overparameterized models, allowing comparison across non-nested models with different numbers of fitted parameters.

The Akaike Information Criterion (AIC), its small-sample correction (AIC<sub>c</sub>), and the



Bayesian Information Criterion (BIC) are defined as:

$$\text{AIC} = n \ln \left( \frac{\text{SSE}}{n} \right) + 2p, \quad (30)$$

$$\text{AIC}_c = \text{AIC} + \frac{2p(p+1)}{n-p-1}, \quad (31)$$

$$\text{BIC} = n \ln \left( \frac{\text{SSE}}{n} \right) + p \ln n, \quad (32)$$

where  $n$  is the number of data points and  $p$  the number of fitted parameters.

The first term in each expression reflects the goodness of fit through the residual variance, while the second term introduces a penalty that increases with model complexity. AIC applies a constant penalty per parameter, whereas BIC imposes a stronger penalty that scales with  $\ln n$ , favoring simpler models as the dataset grows.

For small datasets, which are common in adsorption experiments, the corrected form  $\text{AIC}_c$  is preferred, as it compensates for finite-sample bias and prevents the selection of overly flexible models.  $\text{AIC}_c$  is meaningful only when  $n > p + 1$ ; models violating this condition are rejected as underdetermined. For weighted fits, the corresponding weighted log-likelihood should be used instead of the unweighted SSE expression.

Models are ranked according to their relative information loss, quantified by the difference

$$\Delta \text{AIC}_c = \text{AIC}_{c,i} - \text{AIC}_{c,\min}. \quad (33)$$

The conventional interpretation is that  $\Delta \text{AIC}_c < 2$  indicates substantial support for multiple models,  $4 < \Delta \text{AIC}_c < 7$  indicates considerably less support for the higher-AIC model, and  $\Delta \text{AIC}_c > 10$  indicates essentially no support.

When several models fall within  $\Delta \text{AIC}_c < 2$ , model discrimination becomes ambiguous. In such cases, Akaike weights (Eq. S-132) can be used to estimate the relative likelihood of each model given the data. These weights provide a probabilistic interpretation of model support and can be used to perform model-averaged predictions, thereby reducing sensitivity

375 to model selection.

376 Importantly, the use of information criteria does not imply that one model is “true,” but  
 377 rather identifies the model that achieves the best trade-off between accuracy and parsimony  
 378 within the candidate set. In the context of adsorption isotherms, this is essential to avoid  
 379 overinterpreting parameters obtained from overly flexible models that fit noise rather than  
 380 underlying adsorption behavior.

### 381 5.3 F-test for nested models

382 When a simpler model is nested within a more complex one (e.g. Langmuir  $\subset$  Sips, Langmuir  
 383  $\subset$  Double Langmuir), the F-test determines whether the additional parameters provide a  
 384 statistically significant improvement:

$$F = \frac{(\text{SSE}_1 - \text{SSE}_2)/(p_2 - p_1)}{\text{SSE}_2/(n - p_2)}. \quad (34)$$

385 The F-test is treated as an auxiliary criterion because nested comparisons can be delicate  
 386 when the additional parameter lies near a boundary, such as the Sips limit  $m = 1$ . Bootstrap  
 387 identifiability and physical admissibility should confirm any F-test-based decision. Beyond  
 388 the F-test, physically motivated criteria can guide the choice between nested alternatives:

- 389 • **Langmuir vs. Sips:** Accept Langmuir if the Sips-derived intrinsic heterogeneity is  
 390 negligible,  $\sigma_H^{(\text{Sips})} = (\pi/\sqrt{3})RT\sqrt{1/m^2 - 1} \leq \alpha RT$ , or if  $m$  is statistically indistin-  
 391 guishable from 1 within uncertainty (Eq. S-139).
- 392 • **Langmuir vs. Double Langmuir:** Accept Double Langmuir if the two site classes  
 393 are resolvable,  $\Delta E_{12} = RT|\ln(K_1/K_2)| > \beta RT$  with  $\beta \approx 2\text{--}3$  (Eq. S-140).
- 394 • **Sips vs. Double Langmuir:** Prefer Sips when the site-energy separation is smaller  
 395 than the distribution width,  $\Delta E_{12} < 2\sigma_H^{(\text{Sips})}$  (Eq. S-141).

396 The overlap coefficient (OVL) (Eq. S-136) provides a complementary, distribution-level sim-  
 397 ilarity metric. A complete decision tree integrating all these criteria is given in Figure S-1.

## 398 5.4 Downstream descriptor use: $k$ -Nearest Neighbours classifica- 399 tion

400 The KNN and SVM examples are intended as downstream demonstrations of how the stan-  
 401 dardized descriptor table can be used once a sufficiently large labelled dataset exists. They  
 402 are not used here to claim predictive performance of a trained classifier. Once the stan-  
 403 dardized descriptors  $\{Q_{\max}, \ln K_{\text{aff}}^*, \sigma_H\}$  (Eq. (11)) have been extracted for a collection of  
 404 adsorbent–adsorbate systems, materials can be classified in this descriptor space without  
 405 imposing a parametric model on the group boundaries. The  $k$ -Nearest Neighbours (KNN)  
 406 algorithm assigns a query material to the class most common among its  $k$  closest neighbours:

$$\hat{y}(\mathbf{x}_q) = \arg \max_c \sum_{i \in \mathcal{N}_k(\mathbf{x}_q)} \mathbf{1}[y_i = c], \quad (35)$$

407 where  $\mathcal{N}_k(\mathbf{x}_q)$  denotes the set of  $k$  nearest neighbours of the query point  $\mathbf{x}_q$  in the standardised  
 408 descriptor space, and  $\mathbf{1}[\cdot]$  is the indicator function. Because KNN is non-parametric and  
 409 requires no assumption about group geometry, it is well suited for exploratory classification  
 410 tasks such as:

- 411 • **Performance grouping:** sorting adsorbents into high-, medium-, and low-capacity  
 412 tiers based on  $(Q_{\max}, \ln K_{\text{aff}}^*)$ .
- 413 • **Material family identification:** given a reference library, predicting whether a new  
 414 material behaves like an activated carbon, zeolite, MOF, or biochar.
- 415 • **Pollutant selectivity:** classifying adsorbent–adsorbate pairs by target contaminant  
 416 class (e.g. heavy metals, dyes, pharmaceuticals, PFAS).

- **Homogeneity screening:** separating materials with near-homogeneous surfaces ( $\sigma_H \approx 0$ ) from those with broad energy distributions ( $\sigma_H \gg RT$ ).

Figure 4 illustrates the first of these tasks for three synthetic performance groups projected onto the  $(Q_{\max}, \ln K_{\text{aff}}^*)$  plane.

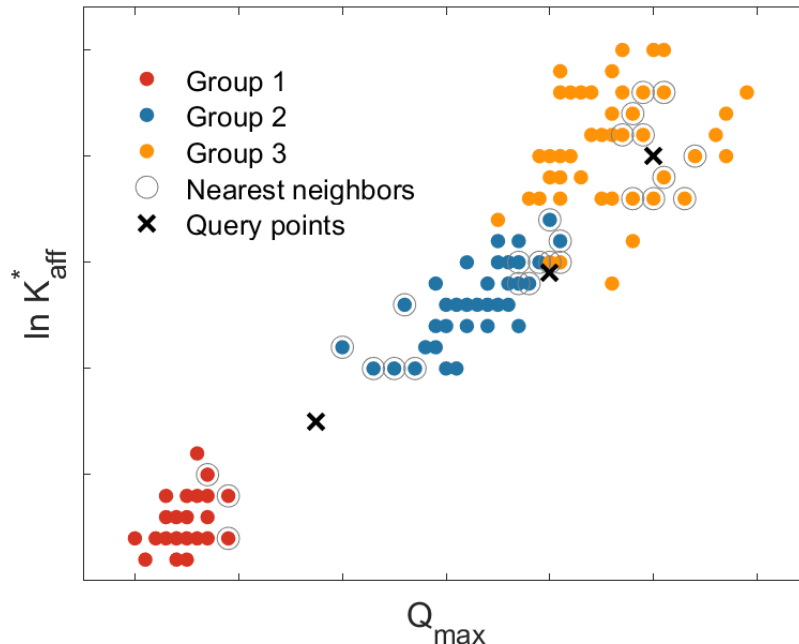


Figure 4:  $k$ -Nearest Neighbours classification of adsorbent materials in the standardized descriptor space  $(Q_{\max}, \ln K_{\text{aff}}^*)$ . Three performance groups are shown: Group 1 (low capacity), Group 2 (intermediate), and Group 3 (high capacity). Query points ( $\times$ ) are classified by majority vote among their  $k = 10$  nearest neighbours (grey circles). **Note:** These synthetic data are shown only to illustrate the classification workflow and should not be interpreted as classifier validation

## 5.5 Material classification by Support Vector Machines

When the classification task requires a well-defined decision boundary, a Support Vector Machine (SVM) finds the maximum-margin separating hyperplane in descriptor space:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C_{\text{reg}} \sum_i \xi_i, \quad y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 - \xi_i, \quad (36)$$

where  $\mathbf{w}$  is the normal to the hyperplane,  $b$  is the bias,  $\xi_i$  are slack variables for misclassified points, and  $C_{\text{reg}}$  controls the trade-off between margin width and classification error. The support vectors—the data points that lie on or within the margin—define the boundary entirely. Figure 5a shows a linear SVM separating two classes of adsorbents in the  $(Q_{\text{max}}, \ln K_{\text{aff}}^*)$  plane (Eq. (11)).

When the full harmonized descriptor set  $\{Q_{\text{max}}, \ln K_{\text{aff}}^*, \sigma_H\}$  is used, the SVM operates in three dimensions and the decision boundary becomes a *plane* (Figure 5b). For non-linearly separable data an RBF kernel  $\mathcal{K}(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2)$  replaces the linear kernel, producing curved decision surfaces that can capture complex group boundaries while still being defined entirely by the support vectors.

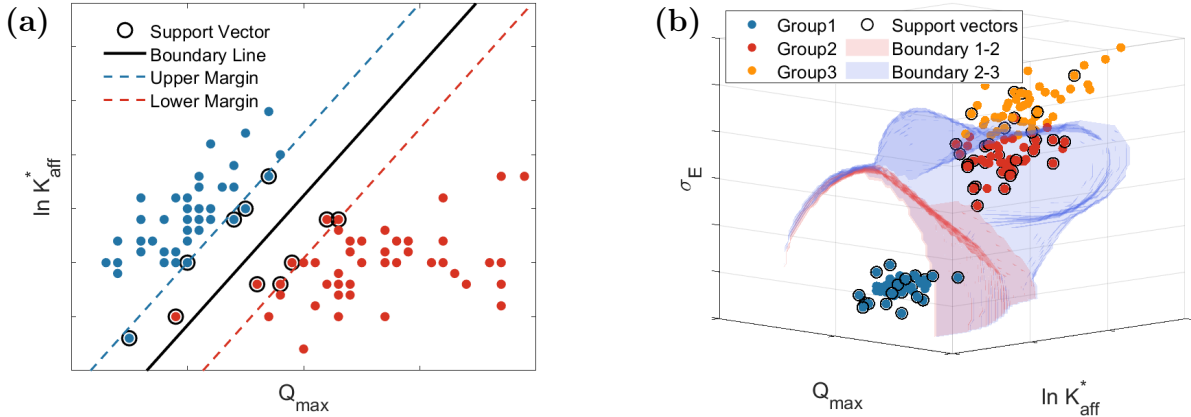


Figure 5: Support Vector Machine classification of adsorbent materials. (a) Two-dimensional projection onto the  $(Q_{\text{max}}, \ln K_{\text{aff}}^*)$  plane: the solid line is the decision boundary  $\mathbf{w} \cdot \mathbf{x} + b = 0$ ; dashed lines indicate the margin boundaries; support vectors are circled. (b) Three-dimensional descriptor space  $(Q_{\text{max}}, \ln K_{\text{aff}}^*, \sigma_H)$ : decision boundaries are now planes (or curved surfaces when an RBF kernel is used). Three performance groups and two inter-group boundaries are shown; support vectors are marked with black circles. **Note:** These synthetic data are shown only to illustrate the classification workflow and should not be interpreted as classifier validation

Categorical variables—such as material type (e.g. activated carbon, zeolite, MOF), best-fit isotherm model, or synthesis route—can be incorporated into the SVM framework through one-hot encoding. Each category is expanded into binary indicator features that enter  $\mathbf{x}_i$  alongside the continuous descriptors  $(Q_{\text{max}}, \ln K_{\text{aff}}^*, \sigma_H)$ . In the resulting augmented feature

space the SVM decision boundary is no longer a plane but a *hyperplane* whose dimensionality equals the total number of features minus one. For a problem with  $p$  continuous descriptors and  $q$  one-hot columns the separating surface lives in  $\mathbb{R}^{p+q}$  and its projection onto any two-dimensional descriptor pair produces the familiar boundary visualised in Figure 5a. Typical SVM classification tasks in adsorption studies include:

- **Binary screening:** suitable vs. unsuitable adsorbents for a given application (e.g. materials exceeding a regulatory removal threshold for drinking water).
- **Model-type separation:** distinguishing systems that are best described by a Langmuir-type isotherm ( $\sigma_H \approx 0$ ) from those requiring a heterogeneous model (Sips, Tóth).
- **Material-class boundaries:** separating material families (activated carbon vs. zeolite vs. MOF) in the descriptor space, using one-hot encoded material type as additional features.
- **Multi-class ranking:** extending the binary SVM via one-vs-one or one-vs-all strategies to classify adsorbents into three or more performance groups (as in Figure 5b).

Both KNN and SVM operate on the same standardized descriptor table (Eq. S-191) with Euclidean distance (Eq. S-193). Ward’s hierarchical clustering (Eq. S-199) provides an unsupervised alternative when class labels are unknown. The key advantage is that all three methods operate on model-harmonized descriptors, ensuring that the grouping reflects intrinsic adsorption performance rather than artefacts of different fitting models. The full mathematical details of all clustering and classification methods are given in Section S-5.

## 6 Worked example: from fits to a reported fingerprint

A practical reporting workflow can be summarized as follows. After fitting the candidate models, models that fail convergence, physical admissibility, or parameter identifiability checks are marked as diagnostic rather than accepted models. Among the remaining models,

information criteria and nested-model tests are used to decide whether the extra parameters are justified by the data. The final reported fingerprint is then extracted only from the selected physically admissible model:

$$\{Q_{\max}, K_{\text{aff}}^*, E_{\text{ads}}, \sigma_H\}.$$

Table 5: Reader-facing example of model triage before descriptor reporting. The numbers are illustrative; the logic, not the synthetic values, is the point of the table.

| Model      | Converged | Physical      | Identifiable         | Reporting status                                         |
|------------|-----------|---------------|----------------------|----------------------------------------------------------|
| Langmuir   | yes       | yes           | yes                  | admissible baseline                                      |
| Sips       | yes       | yes           | yes                  | selected if $\Delta\text{AIC}_c$ justifies heterogeneity |
| Freundlich | yes       | window-only   | no finite $Q_{\max}$ | diagnostic fit, no full fingerprint                      |
| BET        | yes       | range-limited | conditional          | report only within validated regime                      |

This example is intentionally simple. It shows why a visually good fit is not sufficient by itself: a model can be useful for interpolation while still being unsuitable for reporting the complete thermodynamic fingerprint.

## 7 Recommended reporting guidelines

The following guidelines distill the preceding analysis into a practical reporting framework. Their goal is not to prescribe a single isotherm model but to ensure that, regardless of which model is selected, the published results are sufficient for cross-study comparison.

### 7.1 Minimum reporting set

For every adsorbent–adsorbate system analysed, the following quantities should be reported:

1. **Experimental conditions:** temperature  $T$ , pH, ionic strength, and the declared standard-state concentration  $C^0$  (with units).

2. **Model fit:** best-fit model name, all fitted parameters  $\theta$  with units and confidence intervals, and at least SSE,  $R^2$ , ARED, and  $AIC_c$ .
3. **Reported descriptors:**  $Q_{\max}$  (mg g<sup>-1</sup>),  $K_{\text{aff}}^*$  (dimensionless),  $E_{\text{ads}}$  (kJ mol<sup>-1</sup>), and  $\sigma_H$  (kJ mol<sup>-1</sup>).
4. **Comparison metrics** (when applicable): OVL between competing model distributions and/or cluster labels from KNN/SVM classification.

The complete minimum reporting table is given in Table S-3; the extraction formulas for each model are collected in Table S-4.

## 7.2 Recommended analysis workflow

We recommend a seven-step pipeline:

1. **Data collection.** Record  $q_e$ - $C_e$  pairs at constant  $T$  and pH, with at least  $n \geq 8$  points spanning the linear, transition, and plateau regions.
2. **Non-linear fitting.** Fit all models as a diagnostic library, but interpret only the models that are physically admissible, statistically supported, and identifiable from the available data. (Section 5.1).
3. **Error metrics.** Compute SSE,  $R^2$ ,  $R_{\text{adj}}^2$ , ARED, and HYBRID for each model (Section S-2.1).
4. **Model selection.** Rank by  $AIC_c$ ; apply F-tests for nested pairs; apply physical decision criteria (Section 5.3, Figure S-1).
5. **Descriptor extraction.** From the selected model, compute  $Q_{\max}$ ,  $K_{\text{aff}}^*$ ,  $E_{\text{ads}}$ , and  $\sigma_H$  (Table 2).
6. **Statistical comparison.** Compute OVL between energy distributions of alternative models or between different materials (Section S-3.2).



7. **Classification.** Map materials into the standardised descriptor space and apply clustering or supervised classification (Section 5.4, Section 5.5).

### 7.3 Practical recommendations

- **Never linearise.** Linearised isotherm forms distort the error structure and bias the fitted parameters. Always use non-linear regression.
- **Do not report  $Q_{\max}$  from the Freundlich equation.** The Freundlich isotherm has no saturation; any “capacity” derived from it is a concentration-window artefact.
- **Always declare  $C^0$ .** The numerical value of  $E_{\text{ads}}$  depends on  $C^0$ ; omitting it renders the result non-reproducible. Differences  $\Delta E_{\text{ads}}$  between materials measured under the same  $C^0$  are invariant.
- **Report confidence intervals.** Bootstrap or Monte Carlo uncertainties on  $Q_{\max}$ ,  $E_{\text{ads}}$ , and  $\sigma_H$  should accompany every table entry.
- **Use  $\sigma_H$  to assess model necessity.** If the best 3-parameter model yields only weak heterogeneity, e.g.  $\sigma_H \leq 2RT$  or, for Sips,  $m$  remains close to 1 within uncertainty, the Langmuir model is likely sufficient and the extra parameter is not justified (Eq. S-139).
- **Prefer the simplest adequate model.** Among models with  $\Delta\text{AIC}_c < 2$ , report the one with the fewest parameters.

## 8 Conclusions

Adsorption isotherms remain indispensable tools for evaluating materials for water-quality applications, but their fitted parameters cannot be treated as universally interchangeable quantities. This work shows that many common comparisons in the adsorption literature mix parameters with different dimensions, different limiting behavior, and different physical

521 meaning. In particular, a maximum capacity should only be reported when the selected  
 522 model contains a finite saturation plateau. Non-saturating models such as Freundlich may  
 523 be useful empirical descriptions over a limited concentration range, but they do not define  
 524 an intrinsic  $Q_{\max}$  and should not be used to rank materials by maximum uptake.

525 To address this problem, we proposed a unified reporting framework that separates model  
 526 fitting from descriptor extraction. The final result of an isotherm analysis should be reduced,  
 527 whenever possible, to a common descriptor set:

$$\{Q_{\max}, K_{\text{aff}}^*, E_{\text{ads}}, \sigma_H\}.$$

528 Here,  $Q_{\max}$  describes the finite adsorption capacity,  $K_{\text{aff}}^*$  places affinity on a dimensionless  
 529 standard-state scale,  $E_{\text{ads}}$  expresses this affinity as an adsorption-energy descriptor, and  $\sigma_H$   
 530 quantifies intrinsic surface heterogeneity. This fingerprint allows adsorption data obtained  
 531 from different isotherm models to be compared in a physically meaningful way, provided  
 532 that the standard-state concentration  $C^0$ , experimental conditions, fitted parameters, and  
 533 uncertainty metrics are explicitly reported.

534 The framework also provides a practical route for model selection. Rather than relying  
 535 on  $R^2$  alone, the proposed workflow combines non-linear regression, residual-based metrics,  
 536 information criteria, nested-model tests, and physical validation based on limiting behavior  
 537 and energy-distribution families. This approach helps distinguish cases where Langmuir is  
 538 sufficient from cases where heterogeneous models such as Sips, T $\tilde{\text{A}}$ <sup>3</sup>th, UNILAN, Jovanovic,  
 539 or double Langmuir are genuinely justified. It also prevents over-interpretation of small  
 540 statistical improvements that do not correspond to physically meaningful differences in ad-  
 541 sorption behavior.

542 Finally, the descriptor-based structure of the framework makes adsorption data more  
 543 suitable for cross-study comparison, uncertainty analysis, and machine-learning classifica-  
 544 tion. By converting model-specific fitting results into a common set of thermodynamic and

545 heterogeneity descriptors, adsorption studies can move from isolated parameter tables to  
546 ward reproducible structure–performance relationships across adsorbents, contaminants, and  
547 water matrices. The proposed guidelines are therefore not intended to impose a single pre-  
548 ferred isotherm model, but to ensure that whichever model is used, the reported quantities  
549 remain transparent, comparable, and physically interpretable.

## 550 Acknowledgement

551 This work was partially supported by the J-PEAKS program of JSPS.

## 552 Supporting Information Available

## 553 References

- 554 (1) Progress on household drinking water, sanitation and hygiene 2000-2022: Spe-  
555 cial focus on gender - UNICEF DATA. [https://data.unicef.org/resources/  
556 jmp-report-2023/](https://data.unicef.org/resources/jmp-report-2023/).
- 557 (2) The United Nations World Water Development Report 2023: partnerships and co-  
558 operation for water - UNESCO Digital Library. [https://unesdoc.unesco.org/ark:  
559 /48223/pf0000384655](https://unesdoc.unesco.org/ark:/48223/pf0000384655).
- 560 (3) Mekonnen, M. M.; Hoekstra, A. Y. Four billion people facing severe water scarcity.  
561 *Science Advances* **2016**, *2*, e1500323.
- 562 (4) Vörösmarty, C. J.; McIntyre, P. B.; Gessner, M. O.; Dudgeon, D.; Prusevich, A.;  
563 Green, P.; Glidden, S.; Bunn, S. E.; Sullivan, C. A.; Liermann, C. R.; Davies, P. M.  
564 Global threats to human water security and river biodiversity. *Nature* **2010**, *467*, 555–  
565 561.

- 566 (5) Blueprint for Acceleration: Sustainable Development Goal 6 Synthesis Re-  
567 port on Water and Sanitation 2023. [https://www.unwater.org/publications/](https://www.unwater.org/publications/sdg-6-synthesis-report-2023)  
568 [sdg-6-synthesis-report-2023](https://www.unwater.org/publications/sdg-6-synthesis-report-2023).
- 569 (6) Crini, G. Non-conventional low-cost adsorbents for dye removal: A review. *Bioresource*  
570 *Technology* **2006**, *97*, 1061–1085.
- 571 (7) Fu, F.; Wang, Q. Removal of heavy metal ions from wastewaters: A review. *Journal of*  
572 *Environmental Management* **2011**, *92*, 407–418.
- 573 (8) Ahmaruzzaman, M. Industrial wastes as low-cost potential adsorbents for the treatment  
574 of wastewater laden with heavy metals. *Advances in Colloid and Interface Science* **2011**,  
575 *166*, 36–59.
- 576 (9) aus der Beek, T.; Weber, F.; Bergmann, A.; Hickmann, S.; Ebert, I.; Hein, A.;  
577 Küster, A. Pharmaceuticals in the environment—Global occurrences and perspectives.  
578 *Environmental Toxicology and Chemistry* **2016**, *35*, 823–835.
- 579 (10) Thacharodi, A.; Hassan, S.; Hegde, T. A.; Thacharodi, D. D.; Brindhadevi, K.;  
580 Pugazhendhi, A. Water a major source of endocrine-disrupting chemicals: An overview  
581 on the occurrence, implications on human health and bioremediation strategies. *Envi-*  
582 *ronmental Research* **2023**, *231*, 116097.
- 583 (11) Li, J.; Liu, H.; Paul Chen, J. Microplastics in freshwater systems: A review on occur-  
584 rence, environmental effects, and methods for microplastics detection. *Water Research*  
585 **2018**, *137*, 362–374.
- 586 (12) Rahman, M. F.; Peldszus, S.; Anderson, W. B. Behaviour and fate of perfluoroalkyl  
587 and polyfluoroalkyl substances (PFASs) in drinking water treatment: A review. *Water*  
588 *Research* **2014**, *50*, 318–340.

- (13) Buck, R. C.; Franklin, J.; Berger, U.; Conder, J. M.; Cousins, I. T.; de Voogt, P.; Jensen, A. A.; Kannan, K.; Mabury, S. A.; van Leeuwen, S. P. Perfluoroalkyl and polyfluoroalkyl substances in the environment: Terminology, classification, and origins. *Integrated Environmental Assessment and Management* **2011**, 7, 513–541.
- (14) Glüge, J.; Scheringer, M.; Cousins, I. T.; DeWitt, J. C.; Goldenman, G.; Herzke, D.; Lohmann, R.; Ng, C. A.; Trier, X.; Wang, Z. An overview of the uses of per- and polyfluoroalkyl substances (PFAS). *Environmental Science: Processes & Impacts* **2020**, 22, 2345–2373.
- (15) US EPA, O. PFAS Strategic Roadmap: EPA’s Commitments to Action 2021-2024. 2021; <https://www.epa.gov/pfas/pfas-strategic-roadmap-epas-commitments-action-2021-2024>.
- (16) Foo, K. Y.; Hameed, B. H. Insights into the modeling of adsorption isotherm systems. *Chemical Engineering Journal* **2010**, 156, 2–10.
- (17) Hu, Q.; Lan, R.; He, L.; Liu, H.; Pei, X. A critical review of adsorption isotherm models for aqueous contaminants: Curve characteristics, site energy distribution and common controversies. *Journal of Environmental Management* **2023**, 329, 117104.
- (18) Langmuir, I. THE ADSORPTION OF GASES ON PLANE SURFACES OF GLASS, MICA AND PLATINUM. *Journal of the American Chemical Society* **1918**, 40, 1361–1403.
- (19) Freundlich, H. Über die Adsorption in Lösungen. *Zeitschrift für Physikalische Chemie* **1907**, 57U, 385–470.
- (20) Sips, R. On the Structure of a Catalyst Surface. *The Journal of Chemical Physics* **1948**, 16, 490–495.

- 612 (21) Latour, R. A. The langmuir isotherm: A commonly applied but mis-  
613 leading approach for the analysis of protein adsorption behavior. *Journal*  
614 *of Biomedical Materials Research Part A* **2015**, *103*, 949–958, \_eprint:  
615 <https://onlinelibrary.wiley.com/doi/pdf/10.1002/jbm.a.35235>.
- 616 (22) de Vargas Brião, G.; Hashim, M. A.; Chu, K. H. The Sips isotherm equation: Often  
617 used and sometimes misused. *Separation Science and Technology* **2023**, *58*, 884–892,  
618 \_eprint: <https://doi.org/10.1080/01496395.2023.2167662>.

# 619 TOC Graphic

620

